Full Stack Development with MERN

Project Documentation

1. Introduction

• **Project Title: learnhub: your center for skill enhancement options**

Team Members:
Team Size: 4
Team Leader:

- **Palla Harika**– Full Stack Development (Backend APIs, Database Integration, Project Integration) & Frontend Support & Testing **Team Members:**
- **Nikhitha Nadigatla-**Documentation & Presentation Support
- **Sravanthi Killi**– Frontend Development (UI Components & Design)
- **Kanakasri K S S N-**Project Deployment

2. Project Overview

• **Purpose:**

The purpose of this project is to develop a full-stack web application where it provide practical, job-oriented skills. Helps students gain hands-on experience beyond classroom theory.
Here user can register and log in, enroll in courses, a learning materials

- **Features:**

User Features:

- User Registration & Login
- Available trending courses
- Available explore course
- After login user can access learning materials

Frontend (React)

- Built using React.js
- Component-based architecture
- React Router for navigation
- **Axios** Used to connect frontend with backend APIs
- CSS for styling

Frontend communicates with backend using REST APIs.

Backend (Node.js + Express)

- Node.js - Runs JavaScript on the server.
- MangoDB - Stores user data, course details, and other information

Database (MongoDB)

MongoDB used as NoSQL database.

Collections:

- Users
- Courses
- Registrations

3. Setup Instructions

Prerequisites:

- Node.js
- MongoDB
- VS Code

Installation:

Step 1 :https://drive.google.com/drive/folders/1tDDOZqnWSgRxzXw20gyimiAPJI7WNRaz

Open the link extract files in it.

Step 2: Backend Setup

cd backend

npm install

npm start

Step 3: Frontend Setup

cdfrontend

npm install

npm start

4. Folder Structure:

Frontend

- Src/
- Components/
- App.js
- Index.js

Backend

- Components
- Uploads
- Routers

5. Running the Application

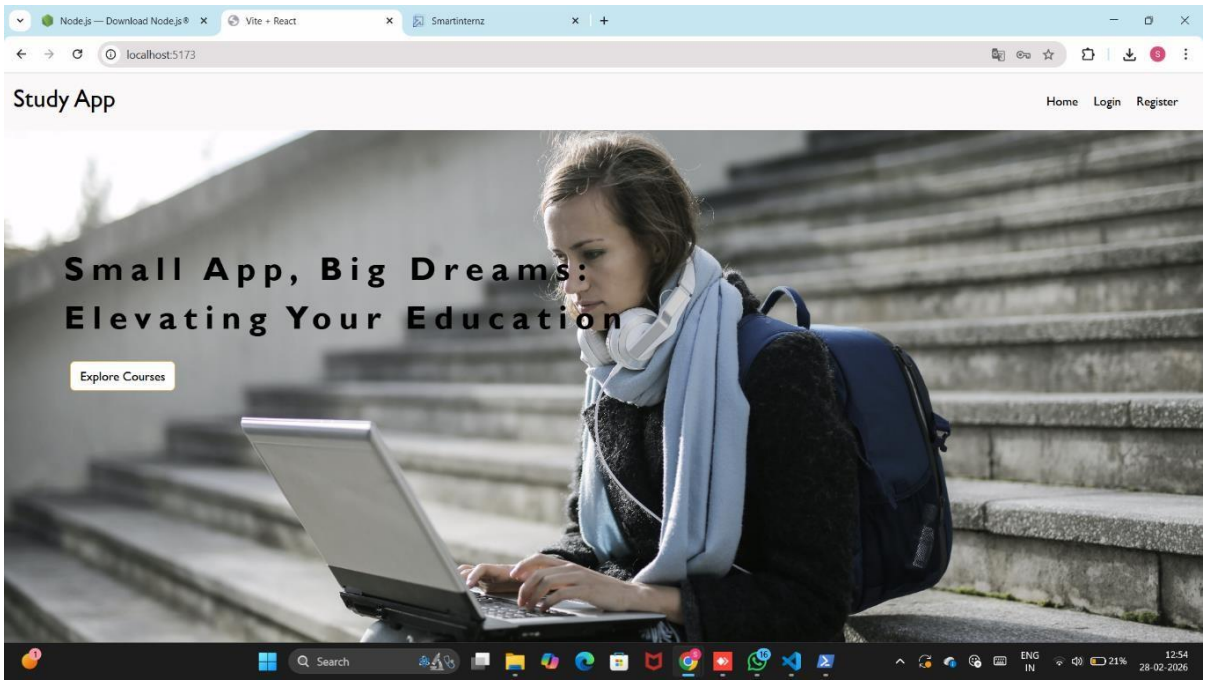
1. Start MongoDB
2. Start Backend:
   npm install
3. Start Frontend:
   npm start

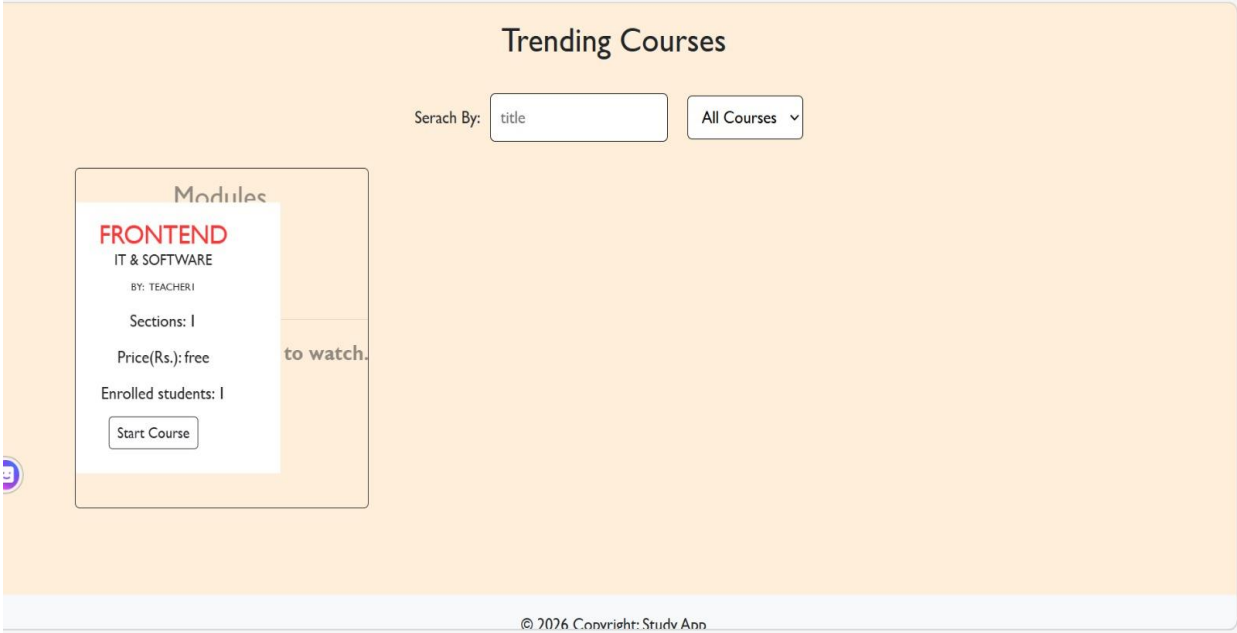
Open: http://localhost:5173/

6. User Interface

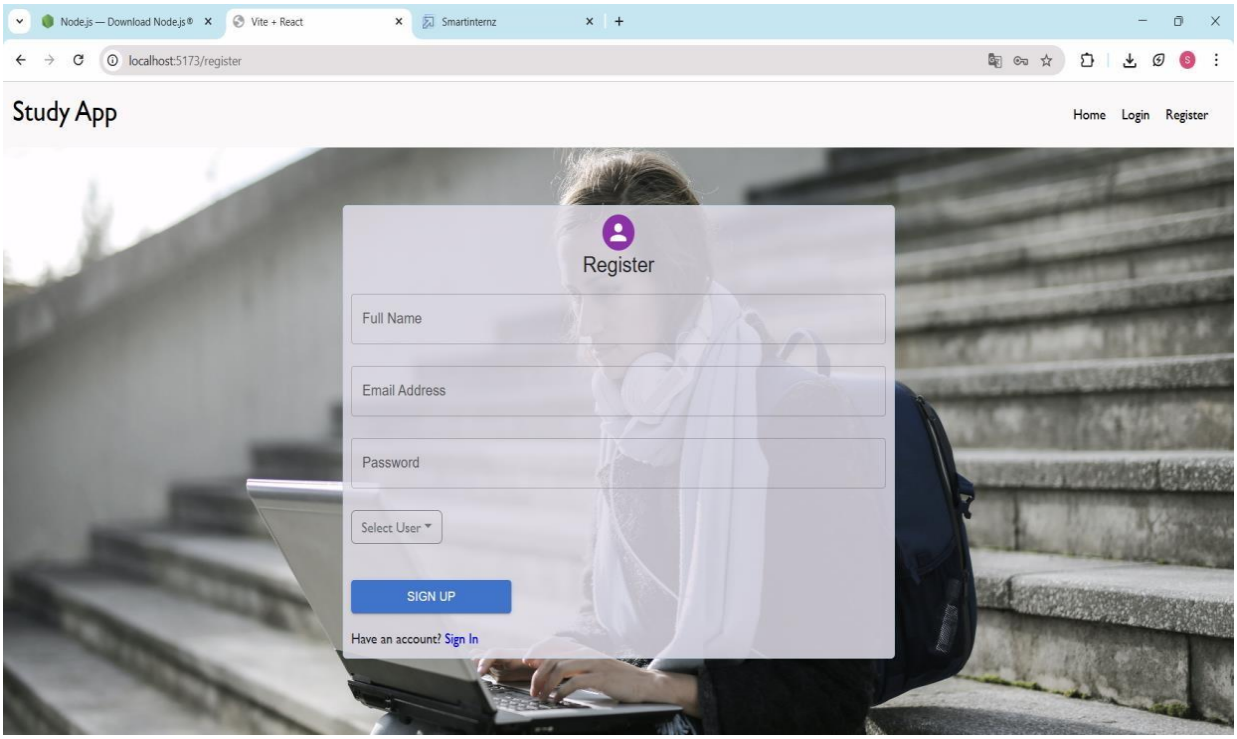
The UI includes:

Home Page:

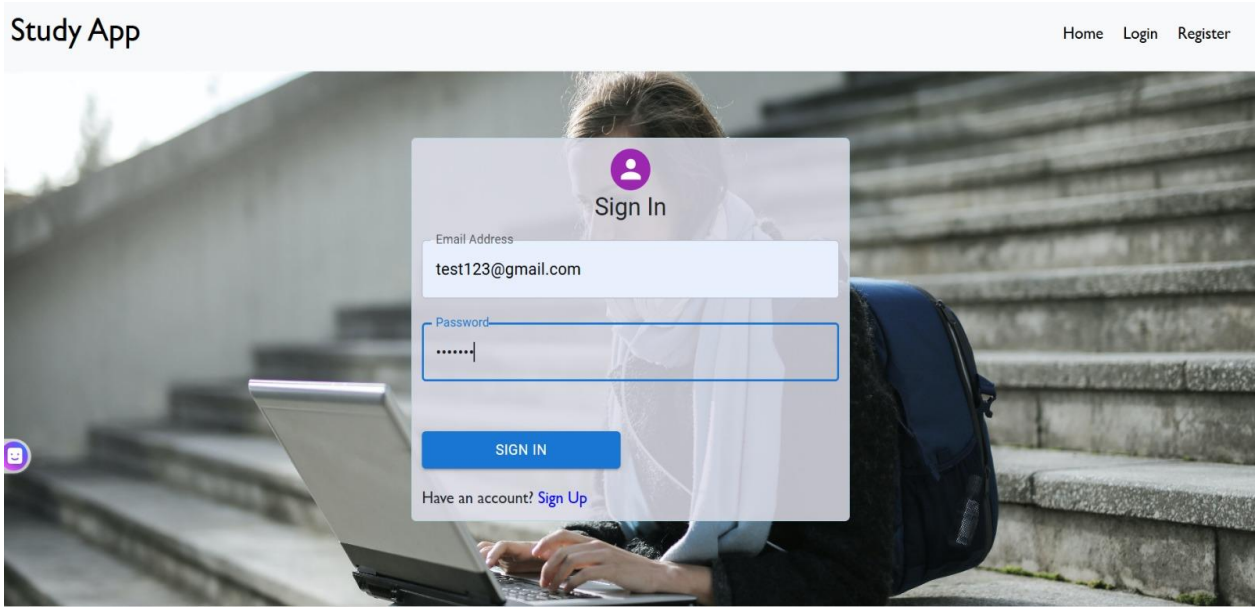




Registration Page



Login Page


Study App
Home
Login
Register
Sign In
Email Address
test123@gmail.com
Password
SIGN IN
Have an account? Sign Up

Study App
Home
Add Course
Hi Test1
Log Out
frontend
Description: html,css Read More
Category: IT & Software
Sections: 1
Enrolled students: 1
Delete
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Study App Home Add Course

Hi Test1 Log Out

frontend

Description: html,css Read More

Category: IT & Software

Sections: 1

Enrolled students: 1

Delete

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Study App Home Add Course

Hi Test1 Log Out

Course Type

Select categories

Course Title

Enter Course Title

Course Educator

Enter Course Educator

Course Price(Rs.)

for free course, enter 0

Course Description

Enter Course description

+ Add Section

Submit

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Study App Home Enrolled Courses

Hi Student1 Log Out

Course ID	Course Name	Cousre Educator	Course Category	Action
69a95f6f8f78b4c06313df8f	frontend	teacher1	IT & Software	GO TO

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Study App Home Enrolled Courses

Hi Student1 Log Out

Welcome to the course: frontend

PLAY VIDEO

DOWNLOAD CERTIFICATE

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1. Testing

- Manual testing performed for all APIs

Verified:

- Registration
- Login
- Add Courses
- View Registrations

Tested using:

Browser

2. Known Issues

- No email notifications
- Basic admin authentication
- UI responsiveness can be improved

3. Future Enhancements

- Add live class feature using WebRTC or third-party APIs.
- Recorded lecture upload system.
- Real-time doubt clarification.
- Timed mock tests.
- Automatic result generation.